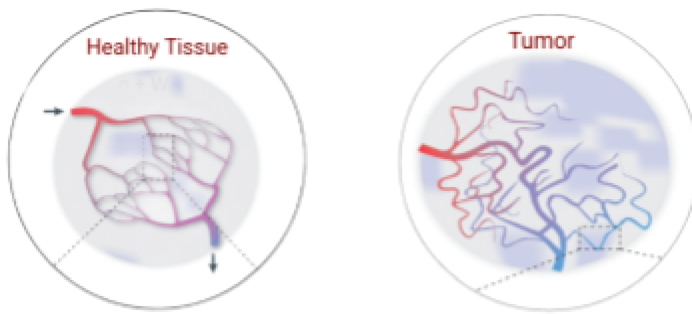

Figure 1: Healthy tissue maintains organized blood flow, whereas tumor vasculature exhibits chaotic pathways.

We partnered with Karczmar Lab, a breast imaging research lab at the University of Chicago, to investigate whether early changes in breast-tumor vasculature could predict which patients will be cancer-free at surgery. Currently, this can only be confirmed after months of treatment. Earlier predictions would allow clinicians to escalate treatment for patients who need it and de-escalate for those likely to respond well.

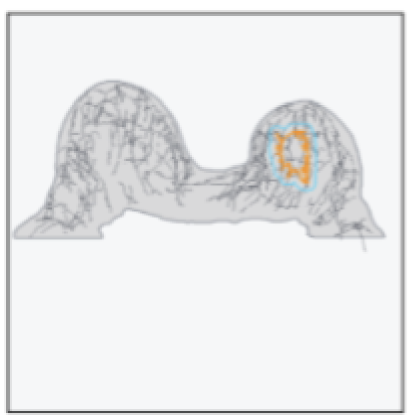

Figure 2: Blood vessel maps extracted from breast MRI scans. These maps reveal the density and organization of vessels surrounding a tumor, which may reflect how aggressively it responds to treatment.

We pursued two complementary approaches to model treatment response. The first focused on tumor appearance. Using MRI scans, we measured characteristics such as shape, texture, and brightness, reflecting tumor structure and blood flow, both of which influence how it grows and how it responds to treatment. The second strategy examined the tumor's environment by analyzing blood vessel density and organization. To support these models, we refined the extraction algorithm to produce cleaner vessel maps and redesigned measurements derived from them. We also processed the full imaging dataset, roughly 8,000 three-dimensional scans, and built a centralized evaluation system enabling fair comparisons across models.

Both tumor appearance and surrounding vasculature carried predictive signals, though accuracy remained moderate. Next steps include integrating vascular and appearance models with clinical data, refining extracted features, and exploring more expressive modeling approaches.